Chapter 8: Ethics — bias, privacy, and intellectual property

Large language models do not emerge from neutral, objective processes. They are built by companies to make money, they are trained on data produced by specific human cultures, and they are fine-tuned by humans.

At every step, choices are made — and those choices have consequences. This chapter examines three of the most important ethical dimensions of LLMs: systemic bias, data privacy, and intellectual property. It also looks at the invisible human workforce that makes these systems possible and the environmental impacts of the technology.

8.1 Systemic bias: what the model learned from us

LLMs learn from human-generated text, and human-generated text reflects human prejudices. Racial stereotypes, gender bias, cultural assumptions, and historical inequalities are all present in the data scraped from the internet, books, and online forums. When the model learns statistical patterns from this data, it internalizes not just human knowledge, but also human biases.

The consequences can be harmful. A study found that when state-of-the-art LLMs were asked to rank resumes, they showed significant racial and gender bias: resumes with names associated with white men were ranked more favorably, while resumes with Black male names were never ranked first across the tested scenarios. The models were not programmed to discriminate — they had simply absorbed text patterns that exist in human society and that in turn affected who they ranked highly.

Image-generating AI systems have shown similar patterns. They typically generate images of 'CEOs' as white men and 'nurses' as women. Efforts to address this bias have not been entirely successful. Google temporarily suspended its AI system's ability to generate images after the system, when asked to generate an image of a 1943 German soldier, returned images of people of color in German uniforms — a historical inaccuracy.

Bias has always existed, but what makes AI bias particularly troubling is the scale at which these systems operate and the credibility people attribute to them. One biased human

hiring manager affects a limited number of candidates. An AI system used by thousands of employers can encode and amplify bias at corporate scale. Moreover, because the output looks authoritative and 'objective' — it comes from a computer, after all — users may be less likely to scrutinize it than they would a human recommendation.

Researchers and companies are working on mitigation strategies: more diverse training data, bias audits, adversarial testing with underrepresented groups, and RLHF specifically designed to reduce discriminatory outputs. Progress is being made, but you should never forget that the results you get from an LLM may have significant biases.

8.2 Data privacy: what happens to what you type?

Every time you type a message into an LLM interface, you are sharing information with the company that operates that service. Depending on the product's terms of service, that information may be logged, reviewed by human annotators for quality control, and in some cases used to train future versions of the model. This creates significant privacy risks that most users do not fully consider.

The risk is especially acute for sensitive information. Employees at companies have been found to have pasted confidential client data, proprietary source code, and internal financial information into ChatGPT while trying to get help with work tasks. In 2023, Samsung engineers inadvertently leaked proprietary semiconductor design information by uploading it to ChatGPT for debugging help.

For individuals, the risks are more personal. People have shared medical symptoms, financial struggles, relationship problems, and mental health concerns with AI chatbots, sometimes treating them like therapists or confessors. The data shared in these conversations may be retained and trained on and may be visible to company employees. Even when companies offer opt-out options for training data use, it is often unclear what really happens to the data.

Practical guidance: treat an LLM conversation as a semi-public document. Do not enter anything you would not be comfortable sharing with the company's employees or, potentially, the public. Use privacy-focused settings when available, and consult your

organization's AI usage policies before using commercial LLM tools for sensitive professional tasks.

One particularly harmful aspect involves agentic AI's access to your private documents. To work well, these systems need access to your email, text messages, contacts, and other private parts of your life. Imagine you and a friend have a text exchange about their medical condition. Your friend might think it's a private conversation, but if you've given an AI system access to your messages, it can learn about your friend's health issues. Your friend might be justifiably furious that you let this happen, even if you didn't realize it was happening.

What can you do about this risk? Be careful about what permissions you give to agentic AI and be willing to accept the risk of giving it a lot of access. Also, realize that your friends might be using agentic AI and that everything you communicate with them could be hoovered up the agent. It is a strange new world.

8.3 Copyright and intellectual property: who owns what the AI learned?

LLMs are trained on text and other creative works produced by human writers, artists, journalists, and programmers. Much of this work is protected by copyright. The fundamental question of whether training an AI model on copyrighted works without permission or compensation constitutes infringement has generated a number of lawsuits in U.S. federal courts, with no definitive resolution in sight.

One high-profile case is The New York Times v. OpenAI and Microsoft, filed in December 2023. The Times alleged that OpenAI incorporated millions of its copyrighted articles into training data without permission, creating a system that can reproduce Times content closely enough to substitute for it. The music industry has filed similar suits — the RIAA brought copyright infringement claims in 2024 against companies whose models were trained on copyrighted recordings. Visual artists have sued image-generation companies, arguing that tools like Stable Diffusion and Midjourney were trained on billions of images scraped without consent.

The legal debate centers on the doctrine of *fair use*, which allows limited use of copyrighted material without permission for purposes like commentary, education, and transformation. AI companies argue that training on copyrighted data is *transformative* — the model does not copy the work, it learns patterns from it, similar to how a human author learns by reading widely. In June 2025, a federal judge agreed. However, other courts have reached different conclusions, and the legal landscape remains unsettled.

Beyond the legal question is an ethical one. Even if training on copyrighted works is legally fair, is it morally fair? A novelist who spent years writing a book receives no compensation when that book is used to train a model that can then produce novels in a similar style — and potentially compete with human authors in the marketplace.

8.4 The human cost: the invisible workforce behind AI

The polished, helpful AI assistants that millions of people use every day would not exist without a largely invisible workforce of human data labelers — people who evaluate model outputs, rank responses, and flag harmful content so that models can be trained to avoid it. This is the human side of RLHF, and the conditions under which this work is performed raise serious ethical questions.

A 2023 TIME magazine investigation revealed that OpenAI contracted with a firm in Nairobi, Kenya, to hire annotators for approximately $1.32 to $2.00 per hour — well below a living wage even in that context. These workers were asked to read and label some of the most disturbing content imaginable: graphic violence, child abuse material, detailed instructions for self-harm. The work was designed to teach the model what not to say, but the human cost to the workers exposed to this content was severe. Multiple workers reported lasting psychological trauma.

The irony is pointed: the most 'intelligent' and 'safe' AI systems in the world depend on low-wage human labor, much of it performed under psychologically harmful conditions, to make them appear intelligent and safe. When you interact with a guardrailed AI assistant that declines to provide harmful information, that boundary was drawn in part by workers who had to be repeatedly exposed to the very content the model was being trained to avoid.

Awareness of this human cost is a prerequisite for an ethical relationship with AI. Advocacy organizations, labor economists, and tech ethicists are pushing for better pay, psychological support, and labor protections for data workers — but progress has been slow. Some companies have begun paying higher wages and offering mental health resources.

8.5 The environmental cost of AI[1]

Many people are very worried about the environmental impact of AI systems. These arise because the computers that train and run them use substantial electricity. Large data centers contain vast networks of computers that run continuously to respond to prompts, generate images, and process other AI requests. Producing that electricity generally creates greenhouse-gas emissions. And, by increasing demand, it can also raise the price of electricity for all consumers. This is especially worrying given the rapid growth in demand for computing infrastructure worldwide. A 2026 report from Lawrence Berkeley National Laboratory estimates that data centers could use nearly 12% of all U.S. electricity by 2030.

Water is another important, sometimes less visible cost. Data-center servers generate heat, and many facilities use water-based cooling systems to keep equipment from overheating. Water may also be used indirectly: power plants often require water to generate the electricity consumed by data centers, and manufacturing computer chips requires highly purified water. In regions already facing drought or water stress, large facilities can compete with households, farms, and ecosystems for limited local supplies. Some operators are improving efficiency or using recycled water, but the tradeoffs still depend on local conditions.

A brief text exchange with a chatbot may have modest energy and water demands but the cumulative impact can become significant when millions of people use them every day. Individually, the numbers are small. Google calculated in 2025 that a median text prompt to its Gemini chatbot used 0.24 watt-hours of electricity (less than a television uses in nine seconds) and about five drops of water. At those rates, one person's chatbot use has minimal environmental impact.

[1] Some of the underlying points in this section are taken from a Substack post by Zeke Hausfather.

Agentic AI can raise the stakes because it does more than produce a single response. An AI agent may plan a task, call tools, browse or search repeatedly, write and run code, query databases, revise its work, and coordinate with other agents. For complex tasks, it can run for hours. Each step can require another model call and additional computation, so a single request consumes far more electricity and water than a conventional chatbot exchange.

How much more? Climate scientist Zeke Hausfather, a heavy user of AI coding agents, tallied the energy used by his own agents over eight weeks in 2026. His 1,138 typed prompts set off more than 14,000 separate model calls, which processed 3.2 billion tokens. He estimates that this used about 170 kilowatt-hours of data-center electricity, or roughly 150 watt-hours per prompt — around 600 times the energy of a median chatbot prompt. On an average day, his agents used more electricity than two refrigerators. These estimates are uncertain by about a factor of two either way, partly because nobody outside the AI companies knows how much energy a frontier model uses per token.

Why so much? As Chapter 6 explained, an LLM has no memory of its own: each time it responds, it rereads the entire conversation. An agent does the same on every pass through its loop (Chapter 5). Whenever it runs a command, opens a file, or calls a tool, the model rereads everything accumulated so far[2], including every file it has opened and the result of every earlier step. In Hausfather's logs, 96% of the tokens came from this rereading, while the text he actually saw made up less than half a percent.

To put these numbers in perspective, a year of agent use like Hausfather's would consume about 1,100 kilowatt-hours, roughly 10% of the electricity an average American household uses, and emit the equivalent of about 370 kilograms of carbon dioxide on the average U.S. grid. That is about half the emissions of one round-trip economy flight between San Francisco and New York, or roughly 2% of the average American's annual carbon footprint. So even for a heavy user, AI is a modest part of a personal carbon footprint. The concern is that it is a new and rapidly growing source of emissions at a time when emissions need to fall.

[2] Note that most rereads come from a cache that costs roughly a tenth as much energy as fresh input, which is why the energy gap (about 600 times) is far smaller than the token gap (2.9 million tokens per prompt versus about 1,000 for a chat exchange).

There is a strong incentive for the LLM companies to improve the efficiency of their models. Google reports that the energy used by its median Gemini text prompt fell 33-fold in a single year. Yet total AI energy use keeps rising, because cheaper computing invites more use. In any event, the impact can be minimized if we run data centers on clean, climate-safe renewable energy.

8.6 Key takeaways

- LLMs absorb and can amplify human biases present in their training data, producing discriminatory outputs in hiring, image generation, and other applications.
- Sensitive information entered into commercial LLM interfaces may be retained, reviewed by humans, or used in future training — treat these conversations as semi-public.
- More than 50 lawsuits are active over whether training AI on copyrighted works without permission or compensation constitutes infringement — the legal question is unresolved.
- The 'safe' behavior of AI assistants is produced through RLHF carried out by human annotators who are often poorly paid and exposed to severely harmful content.
- Ethical AI use requires awareness of these systemic issues — as consumers, we have a role in advocating for responsible development and deployment.
- AI consumes serious amounts of energy and water. In particular, a single request to an AI agent can use hundreds of times as much energy as a chatbot prompt. We must be cognizant of the environmental impact when using these tools and AI companies need to work to make their models as efficient as possible and to power them with clean electricity.